Northbridge University Endowment: Investment Policy Statement

Prepared by the investment consultant for the Investment Committee · Proposed for adoption · Capital market assumptions as of 30 September 2025 · Fictional client

1. Purpose

This Investment Policy Statement (IPS) sets out how the assets of the Northbridge University Endowment (the Endowment) are invested. It defines the Endowment's objectives and constraints, the strategic asset allocation that the Investment Committee (the Committee) has adopted to meet them, the permitted ranges around that allocation, the policy benchmark against which results are measured, and how the portfolio is rebalanced, monitored and reviewed. It is written to give continuity across changes in Committee membership and staff, and to protect the Endowment from decisions made under short-term market pressure.

The Endowment has $2.0 billion of assets and a perpetual horizon. It supports the University's operating budget through an annual distribution. Its central financial goal is **intergenerational equity**: future students and faculty should receive at least the same real support as current ones, which requires the Endowment to preserve the purchasing power of its assets after spending and costs.

2. Governance and roles

- **Board of Trustees:** approves this IPS and the spending rule; appoints the Committee; receives an annual report.
- **Investment Committee:** meets quarterly; approves the strategic allocation, ranges and benchmark; appoints and dismisses the consultant and investment vehicles; reviews performance and risk each quarter; conducts the annual review in Section 9. Changes to the strategic allocation require a Committee vote.
- **Investment staff (CIO office):** implements the policy within the ranges; executes rebalancing under Section 7 without further approval; reports breaches to the Committee chair within two business days.
- **Investment consultant:** provides capital market assumptions, asset-liability and allocation analysis, performance and risk reporting, and independent advice. The consultant is a fiduciary for its advice.
- **Custodian:** holds the assets, provides independent valuations and transaction records.

3. Objectives

- **Return objective:** a long-term net total return of US inflation + 4.9% a year, which covers spending (4.5%) and costs (0.40%) and so preserves real value. Using the J.P. Morgan 2026 inflation assumption of 2.5%, the nominal hurdle is 7.4% a year (compound).
- **Spending rule:** each year the Endowment distributes 4.5% of the average market value of the trailing 12 quarter-ends. Averaging smooths the budget: a 20% market fall reduces spending gradually over three years.
- **Costs:** all-in investment costs of 0.40% a year, monitored annually.
- **Risk limit:** expected volatility no higher than 12% a year, and no more than a 5% probability of a one-year loss worse than –20%, both measured on the consultant's capital market assumptions.

4. Constraints

- **Liquidity and mandate:** liquid assets only. The Endowment invests through 11 US-listed ETFs (Section 5) and holds no private assets, so it can meet spending and rebalance at any time.
- **Concentration:** no more than 40% in any one ETF.

- **Asset-class limits:** equities (SPY, EFA, EEM, VNQ) 40–70%; Treasuries and TIPS (IEF, TLT, TIP) at least 15%; gold and commodities (GLD, DBC) at most 10%. These limits apply to the strategic (target) allocation; Section 7 explains how they interact with rebalancing.
- **Leverage and derivatives:** none. No short positions.
- **Time horizon:** perpetual; the Committee evaluates the policy over rolling 10-year periods.
- **Tax and legal:** the Endowment is a tax-exempt US educational endowment governed by UPMIFA (the Uniform Prudent Management of Institutional Funds Act), which requires prudent, diversified investing in light of the institution's needs.

5. Strategic asset allocation and ranges

The Committee adopts the **Growth** portfolio. It was chosen from three candidates on the constrained efficient frontier (Conservative 8%, Balanced 10% and Growth 12% expected volatility), using the rule set in advance: the candidate with the highest probability of preserving real value over 20 years that also meets the risk limit. All three candidates met the risk limit, and Growth had the highest probability (55.5%). Appendix A summarises the analysis.

For implementation the frontier weights (EEM 19.9%, TLT 15.0%, HYG 15.1%, VNQ 40.0%, GLD 10.0%) are rounded to whole percentages. Of the roundings that sum to 100% and meet every limit, the one with the highest expected return is adopted. Rounding EEM up to 20% would have lifted expected volatility to just above 12%, so EEM is 19% and HYG 16%. The rounded policy keeps a 55.2% probability of preserving real value.

ETF	Asset class	Target	Lower	Upper
EEM	Emerging market equities	19.0%	15.2%	22.8%
TLT	US Treasuries 20y+	15.0%	12.0%	18.0%
HYG	US high yield	16.0%	12.8%	19.2%
VNQ	US REITs	40.0%	32.0%	40.0%
GLD	Gold	10.0%	8.0%	12.0%
Total		100%		

Table 1. Strategic allocation and permitted ranges (±20% of each target, capped at the 40% single-ETF limit). ETFs with a 0% target are not held.

Expected characteristics (J.P. Morgan 2026 assumptions): arithmetic return 8.5%, compound return of about 7.8%, volatility 11.9%; equities 59%, Treasuries 15%, credit 16%, gold 10%. The expected compound return exceeds the 7.4% hurdle, but only by about 37 basis points.

Concentration warning. The optimizer allocates 0% to US large caps and the maximum 40% to US REITs, because the assumptions give REITs the highest expected return of the 11 asset classes (8.8% compound vs. 6.7% for US large caps). The allocation also sits exactly on three limits (VNQ 40%, Treasuries 15%, gold 10%). Mean-variance optimization concentrates in whichever assets have the highest estimated return. Lowering the REIT assumption by 2 percentage points, all else equal, cuts the REIT weight of the 12%-volatility portfolio from 40% to 9% (Table A3). The Committee should adopt this allocation knowing that it is a large active bet on REITs and emerging markets relative to a market-cap-weighted portfolio.

6. Policy benchmark

The policy benchmark is the strategic allocation in Table 1, invested in the same ETFs at their target weights and rebalanced monthly: 19% EEM, 15% TLT, 16% HYG, 40% VNQ, 10% GLD. Because the benchmark uses the same vehicles, the difference between the Endowment's return and the benchmark measures implementation (drift within ranges, trading costs and timing), not asset allocation. The Committee also monitors two reference measures: the return objective (inflation + 4.9%) to judge whether the policy is meeting its goal, and a 60% MSCI ACWI / 40% Bloomberg US Aggregate portfolio to judge the policy against a simple alternative.

7. Rebalancing

- Staff check the weights at every month-end.
- If any holding is outside its range in Table 1 (±20% of its target, relative), staff rebalance the whole portfolio back to target weights at the next month-end close.
- Cash flows (spending, gifts) are used first to move weights towards targets, reducing trading.
- The asset-class limits in Section 4 govern the target allocation. They are not separate triggers for drifted weights, with one exception: no ETF may exceed 40% for more than one month-end.

Why the limits are not separate triggers. Three targets sit exactly on a limit, so any drift would breach one. On actual ETF returns from May 2007 to September 2026 (233 month-ends), the ±20% bands alone would have triggered 23 rebalances. Treating every limit as a trigger would have triggered 217, which means rebalancing almost every month, and allowing a 1-point tolerance would still have triggered 73. Frequent rebalancing adds cost without improving risk control.

8. Monitoring and reporting

- **Monthly (staff):** market value, weights vs. ranges, returns vs. the policy benchmark, any breaches.
- **Quarterly (Committee):** performance vs. benchmark and objective over 1, 3, 5 and 10 years; risk report (volatility, Value-at-Risk, Expected Shortfall, stress tests); spending calculation; rebalancing log.
- **Annually (Board):** a summary of results against the objective, risk limit, costs and this IPS.
- **Breaches:** any breach of this IPS is reported to the Committee chair within two business days with a plan to correct it.

9. Annual review

The Committee reviews this IPS every year, after the consultant publishes updated capital market assumptions, and whenever the University's circumstances change materially (spending needs, a large gift, a change in the spending rule). The review re-runs the frontier and Monte Carlo analysis with the new assumptions, checks the allocation against the risk limit, reviews costs, and confirms or amends the allocation. The allocation is not changed because of recent performance alone. Any change is recorded with its reasons.

Appendix A. Supporting analysis

A1. Capital market assumptions

J.P. Morgan asset class	ETF	Compound	Arithmetic	Volatility	Hist. arith. 2007–26	Hist. vol. 2007–26
U.S. Large Cap	SPY	6.7%	7.9%	16.5%	11.5%	15.5%
EAFE Equity	EFA	7.5%	8.9%	17.6%	5.9%	17.3%
Emerging Markets Equity	EEM	7.8%	9.7%	20.9%	6.9%	21.0%
U.S. Intermediate Treasuries	IEF	4.0%	4.1%	3.5%	3.1%	6.7%
U.S. Long Treasuries	TLT	4.9%	5.7%	13.0%	3.4%	14.1%
TIPS	TIP	4.3%	4.5%	5.9%	3.3%	5.8%
U.S. Inv Grade Corporate Bonds	LQD	5.2%	5.5%	7.4%	4.0%	8.2%
U.S. High Yield Bonds	HYG	6.1%	6.5%	8.7%	5.1%	10.2%
U.S. REITs	VNQ	8.8%	10.2%	17.4%	7.4%	22.2%
Gold	GLD	5.5%	6.8%	16.7%	10.5%	17.6%
Commodities	DBC	4.6%	6.2%	18.3%	4.1%	19.1%

Table A1. J.P. Morgan Asset Management 2026 Long-Term Capital Market Assumptions (U.S. dollar, 10–15 year horizon, as of 30 September 2025), mapped to the closest ETF, with historical ETF figures for comparison. Inflation assumption: 2.5%. Cash: 3.1%.

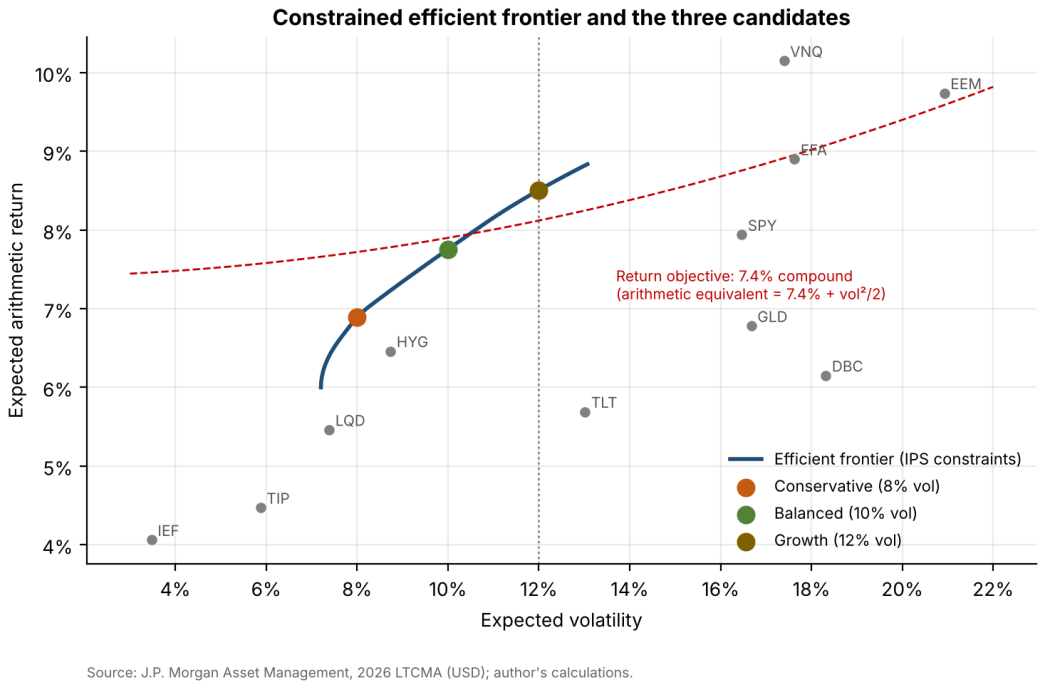


Figure A1. Constrained efficient frontier and the three candidates.

A2. Candidates and Monte Carlo results

	Conservative	Balanced	Growth
SPY	0%	0%	0%
EFA	6%	0%	0%
EEM	5%	6%	20%
IEF	40%	6%	0%
TLT	3%	13%	15%
TIP	0%	0%	0%
LQD	0%	0%	0%
HYG	7%	30%	15%
VNQ	29%	35%	40%
GLD	10%	10%	10%
DBC	0%	0%	0%
Expected return (arith.)	6.9%	7.8%	8.5%
Volatility	8.0%	10.0%	12.0%
P(real value preserved, 20y)	31.6%	47.2%	55.5%
Median real value after 20y	$1.68bn	$1.94bn	$2.15bn
P(drawdown worse than –30%)	21.9%	34.4%	48.2%
P(1-year loss worse than –20%)	0.00%	0.08%	0.34%
Median real spending, year 1	$88.7m	$88.7m	$88.8m
Median real spending, year 20	$73.9m	$83.9m	$92.7m

Table A2. Candidate weights and 10,000-path, 20-year Monte Carlo results. Spending in today's dollars.

Method. Quarterly simulation over 20 years. Asset returns and inflation are drawn jointly from a multivariate lognormal distribution matching the J.P. Morgan arithmetic returns, volatilities and correlations (including inflation). Each quarter the portfolio is rebalanced to target, costs of 0.10% are deducted and spending of 1.125% of the trailing 12-quarter average value is paid. 'Real value preserved' means the market value after 20 years, deflated by simulated inflation, is at least $2.0bn. Drawdowns are measured on quarter-end nominal values after spending. The one-year loss probability uses calendar-year returns after costs and before spending, across all simulated years.

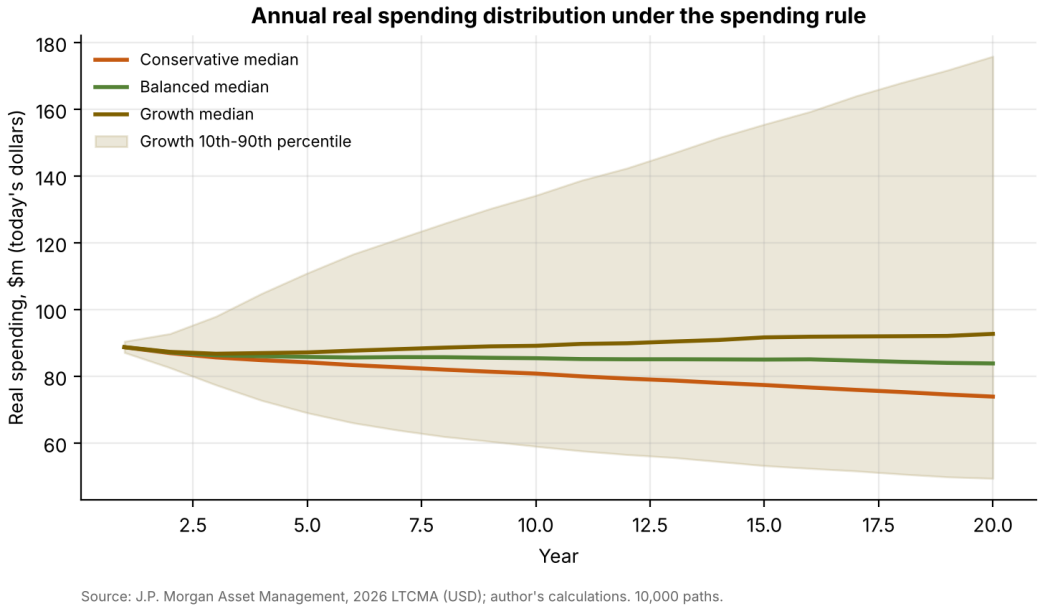


Figure A2. Median annual real spending for each candidate, with the 10th–90th percentile band for Growth.

ETF	Base case	REIT return –1 pt	REIT return –2 pts
EFA	0%	2%	24%
EEM	20%	22%	25%
TLT	15%	15%	20%
HYG	15%	15%	13%
VNQ	40%	36%	9%
GLD	10%	10%	10%

Table A3. Sensitivity of the 12%-volatility frontier portfolio to the REIT expected-return assumption.

A3. Historical cross-check

Applied to actual ETF returns from May 2007 to September 2026 with the band rebalancing policy, the Growth allocation returned 6.0% a year with 14.0% volatility, a worst 12-month return of -37.7% (12 months to February 2009) and a maximum drawdown of -42.6% (trough February 2009). It also lost 21.9% over 12 months in 2022, when stocks and bonds fell together. Realised volatility was higher than the 11.9% assumed, and a –20% one-year loss, which the model treats as a 0.3% event, happened twice in this sample. Normal and lognormal models understate the probability of crises; the Committee should treat the risk limit as a floor on caution, read the risk report's stress tests alongside it, and expect a drawdown of 40% or more in a severe crisis.

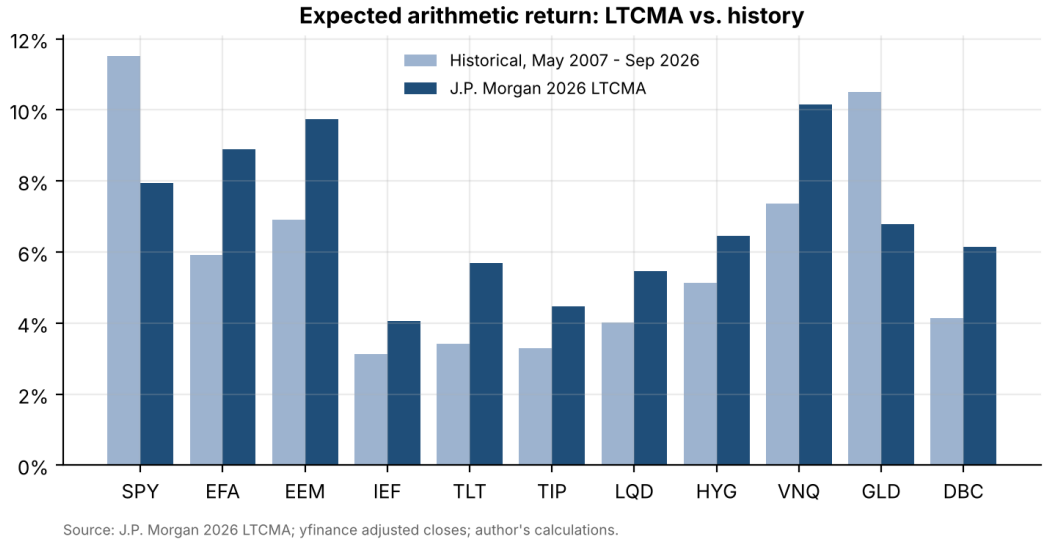


Figure A3. Expected arithmetic returns: J.P. Morgan assumptions vs. 2007–2026 history.